Random Forest Paper Materials

Model Specifics

Model Parameters

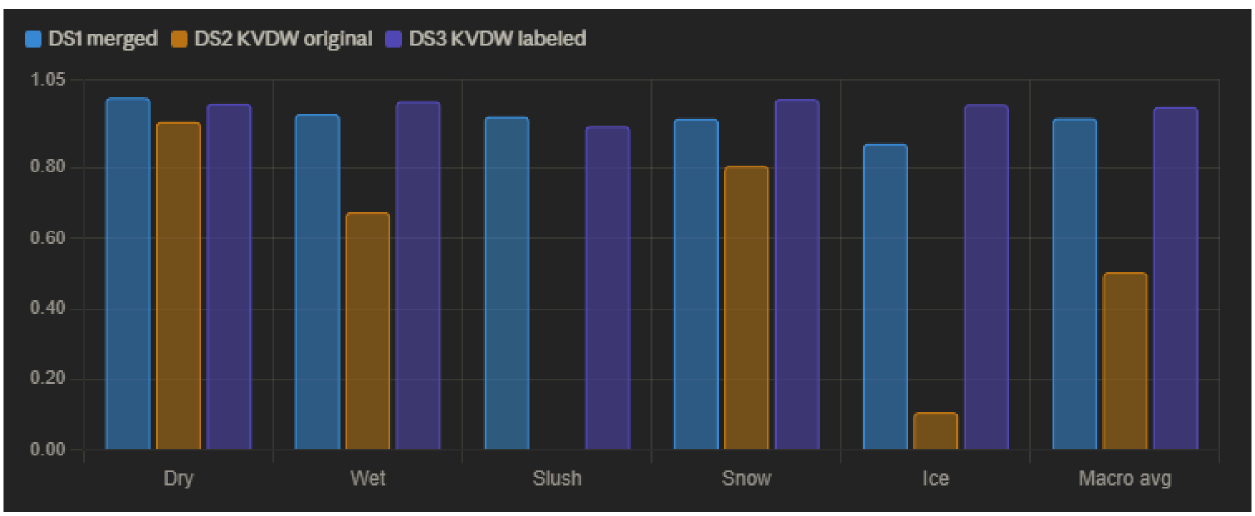
N_estimators: 300
Max_depth: 20
Min_samples_split: 2
Min_samples_leaf: 1
Max_features: "sqrt"
Class_weight: "balanced"
Random_state: 42
N_jobs: -1 (all cores)

Dataset Performance Comparison

KVDW Original (Operator Labeled) dataset
Overall Accuracy: 88.11%

Merged new dataset
Overall Accuracy: 94.2%

KVDW new labeled dataset
Overall Accuracy: 98.87%



Dataset Specifics

Synoptic dataset
File: KVDW.2025-04-25_cleaned_filled_multi.csv

Description: This is the original dataset created with a combination of sensor values from historical Synoptic weather station data as the input and a one hot encoding of the output classes. The output labels are determined using WYDOT operator reports from that same station i.e. I have sensor values at a certain time and also an operator report of wet roads at that same time then that sample is labeled as wet. Finally, due to limited operator labels, the sensor samples with no operator label were predicted based on the learned relationships between sensor values and labels for all operator labeled samples.

Notes:
The visibility input feature was determined to be unimportant and is no longer included in training.

Our dataset
File: merged_dataset_new.csv

Description: This dataset consists of the same environmental sensor values recorded from my setup as well as labeling provided from the SurfaceVue10 sensor that we have. All the data here was collected from the outdoor testbed therefore labels have been thoroughly verified.

Notes:
Labels from the Surfacevue include things like Snow/Frost and Moist but these were grouped into Snow and Wet respectively during training making the output classes simply: Ice, Wet, Dry, Snow, and Slush.

New Synoptic Set
File: KVDW_labeled.csv

Description: This is the same Synoptic data from the original KVDW set but labeled according to rules learned by our created test bed dataset.

Notes:
The labeling here is based solely on the relationships learned from the environmental sensor input values, mean/std voltage is ignored because these measurements are part of the sensor I developed using the reflected light levels.